MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE TSGU (INCLUDING VARIATION TYPE TSGA)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

Sizes 3 to 1/23 inclusive, extruded silicone rubber insulation

First - Copper conductor, uncoated (see table I for size).
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid.
Fourth - Braid covering. (See table I for diameter.) Standard identification code applied by method 1.

Sizes 1/23 to 400 inclusive, silicone rubber-glass tape insulation

First - Copper conductor, uncoated (see table I for size).
Second - Silicone rubber-glass tape insulation (see table I for diameter). Letter identification code applied by method 5.
Third - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either nonfibrous or a combination of nonfibrous and fibrous fillers shall be used to form a firm, well rounded assembly.
Fourth - A binder.

1/ Size 23 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

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Construction variant for type TSGU

Sizes 3 to 1/ 23 inclusive, extruded silicone rubber insulation

Fifth - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either nonfibrous or a combination of nonfibrous and fibrous fillers shall be used to form a firm, well rounded assembly.
Sixth - Binder or combination binder and barrier.
Seventh - Special thermoplastic jacket (gray). (See table I for thickness.) Cable surface marking required.

Sizes 1/ 23 to 400 inclusive, silicone rubber-glass tape insulation

Fifth - Special thermoplastic jacket (gray). (See table I for thickness.) Cable surface marking required.

Construction variant for type TSGA

Fifth - Special thermoplastic jacket (gray). (See table I for thickness.) Cable surface marking required.
Sixth - Braided metal armor and paint.

1/ Size 23 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

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TABLE I. Details.

1/ Size	Conductor size		Type, overall diameter (inches)				Diameter over outer insulation tape nominal (inch)	Diameter over braid covering (max) (inch)	Diameter over insulation (min) (inch)	Jacket thickness nominal (inch)	Conductor resistance per 1000 feet at 25°C (max) (ohms)	Insulation resistance per 1000 feet (min) (megohms)	Cold bending mandrel diameter maximum (inches)	Accelerated service loading amperes
			TSGU		TSGA									
	Navy standard	AWG	min	max	min	max								
3		16(class B)	---	0.411	---	0.461	----	0.130	0.096	0.030	4.18	500	----	----
4		14(class B)	---	.449	---	.499	----	.143	.112	.030	2.68	500	----	----
9		10(class B)	---	.575	---	.625	----	.187	.154	.040	1.06	500	----	----
14		9(class B)	---	.718	---	.768	----	.262	.230	.040	0.8407	500	----	90
23		7(class B)	---	.812	---	.862	0.251	.310	.271	.050	.5284	500	----	115
50		3(class C)	---	.969	---	1.019	.334	----	----	.050	.205	200	12	185
75		1(class C)	---	1.134	---	1.185	.407	----	----	.050	.129	175	14	250
100		0(class D)	---	1.266	---	1.316	.453	----	----	.060	.102	160	16	295
150		000 (class D)	---	1.515	---	1.565	.557	----	----	.060	.0642	135	19	400
200		0000 (class D)	---	1.669	---	1.719	.634	----	----	.060	.0509	125	21	460
300		300 MCM (class D)	---	1.957	---	2.007	.748	----	----	.075	.0360	110	24	590
400	400(127)		---	2.203	---	2.253	.862	----	----	.075	.0273	100	27	750

1/ Type TSGU and TSGA.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	(see table I)
Voltage withstand - volts, root mean square, minimum	
Sizes 3 through 9	3000
Sizes 14 and larger	5000
Insulation resistance - megohms/1000 feet, minimum	(see table I)
Conductor to conductor and conductor to water (TSGU only)	(see table I)
Conductor to conductor and conductor to armor (TSGA only)	(see table I)
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-915, table XXII, for limits of water leakage	No failure
Abrasion resistance - sizes 3 through 23 (extruded silicone rubber) scrapes, minimum	250
Crack resistance - sizes 3 through 23 (extruded silicone rubber)	No damage
Group B:	
Gas flame - (1 hour)	No failure
Drip - at 95 ± 1°C	Zero
Cold bending, cable - (sizes 50 and larger) (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lbf/in^2, minimum	750
Elongation - percent, minimum	125
Jacket	
Tensile strength - lbf/in^2, minimum	2300
Elongation - percent, minimum	250
Group C:	
Physicals (aged)	
Jacket	
Air oven (168 hours at 136°C)	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	50
Permanence of printing (conductor) - sizes 3 through 23 (extruded silicone rubber) cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Armor-conformance to material coverage and construction.	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Cable aging and compatibility (125 ± 5°C)	No failure
Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Accelerated service - sizes 14 and larger (see table I for load current)	No failure
Flame propagation	No failure
Jacket durometer (type A) hardness, minimum	80

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
TSGU-3	2500
TSGU-4 through 23	2000
TSGU-50	1500
TSGU-75 through 150	1000
TSGU-200 through 400	800

Revision letters are not used to denote changes due to the extensiveness of the changes.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
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